Principled Deletion in Content-Addressed Rule Closures

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Version 1.0 — June 2026

Abstract

Content-addressed knowledge stores that grow by least-fixpoint rule closure are append-only by construction: the growth theory of the companion paper [12] shows that conservativity, history independence, and incremental locality all flow from monotonicity, and that the natural non-monotone request already breaks them. Yet redaction obligations — erasure rights, court orders, consent withdrawal, error correction — are real. This paper develops the deletion theory of such stores. The structural key is that content addressing *inscribes derivations*: every generated part names its premises inside its own address preimage, so derivations are unique and the support of a part is a payload-local traversal. From this we prove the **cone theorem**: deleting atoms D from a closed store removes *exactly* the upward cone $\mathrm{Up}(D)$, so epoch re-closure and incremental cone excision coincide bit-for-bit, survivors keep their exact addresses, and the alternative-derivation accounting that dominates classical view maintenance (DRed) vanishes structurally. We then prove what deletion costs: under maximal-parsimony views, deletions *insert* (the same four-atom witness of the companion paper runs in reverse — the retracted address `362db034e4f1`... reappears); and no payload-determined rule system expresses deletion over the plain subset order at all. Two escapes exhaust the options: coordinate (epochs) or extend the order — the two-phase closure over $(\text{adds}, \subseteq) \times (\text{removes}, \supseteq)$, which we prove monotone, confluent under remove-wins, and permanently tombstoning. An erasure analysis separates the policies physically: tombstones retain payloads; excision and epochs erase; address-only tombstones audit deletions at a quantified deniability cost. The lattice-level core (survivor stability, product monotonicity, order-freedom, tombstone permanence) is kernel-checked in the same self-contained constructive Lean 4 development as the growth theory — the first two axiom-free; the derivation-level cone theorem is proved on paper and machine-checked by a 12-item ledger and 600 randomized property instances. Empirically, cone excision beats epoch re-closure by 4.8–$8.3\times$ with bit-identical fingerprints, and on a natural corpus the worst single-atom deletion touches $\sim 10^{80.85}$ parts under complete semantics but exactly 4 materialized parts under the quotient representation — while 58.1% of single deletions trigger a view insertion.

1 Introduction

The companion paper [12] develops the growth theory of content-addressed rule closures: a store whose parts occupy a typed tower $\mathsf{Atom} < \mathsf{Block} < \mathsf{Section} < \mathsf{Root}$, whose addresses are collision-resistant hashes of payloads and child addresses (Merkle-DAG identity [8, 10]), and whose derived content is the least fixpoint $\mathsf{K}(S)$ of finitary, payload-determined, stratified rules over a seed S of atoms. Under those hypotheses (H1–H4 below) growth is conservative, history-independent, incremental, and continuous — and the boundary of that comfort is the CALM boundary [6, 2]: the one natural non-monotone request, *store only maximal bindings*, is impossible for payload-determined rules, and its cost in a content-addressed store is the retraction of named, referenced parts.

Growth-only, however, is a legal and operational fiction. The right to erasure (Regulation (EU) 2016/679, Art. 17 [4]) applies to personal data regardless of the elegance of the data structure that holds it; the tension between erasure and content-addressed immutability is by now a documented regulatory problem [3]. Court orders, consent withdrawal, and plain error correction raise the same demand. Systems answer it today ad hoc: tombstone flags, garbage-collection epochs, key destruction. This paper asks what the closure theory itself says.

Contributions.

1. **Derivation structure** (§3). Content addressing under our rule format is *premise-inscribing*: a generated part's address commits to its full premise list. Consequently derivations are unique (Lemma 2) and the atom support of any part is computable by a local Merkle traversal — no global state, no provenance side-table.
2. **The cone theorem** (§4). $\mathsf{K}(S \setminus D) = \mathsf{K}(S) \setminus \mathrm{Up}(D)$: deletion removes exactly the upward cone of the deleted atoms (Theorem 4). Three corollaries do the practical work: survivors keep their exact addresses (Corollary 5); deletion is computable cone-locally with no re-closure pass — *semi-naive deletion* (Corollary 6); and the alternative-derivation accounting of DRed-style view maintenance [5] vanishes, because the derivation count is structurally one.
3. **What deletion costs** (§5). Under maximal-parsimony serialization, deletions *insert*: the companion paper's witness runs in reverse, and the address retracted by growth reappears under deletion (Theorem 8). More fundamentally, no payload-determined rule system expresses deletion over the plain subset order (Theorem 12); the only escapes are coordination (epochs) or extending the order.
4. **The two-phase closure** (§6). Taking the second escape: over the product order $(\text{adds}, \subseteq) \times (\text{removes}, \supseteq)$, the live store $\mathsf{K}(A \setminus R)$ is monotone (Theorem 13), confluent over operation interleavings under remove-wins (Theorem 14) — and permanently tombstoning (Proposition 15). This is the closure-theoretic content of the 2P-set construction [11], now with derived knowledge on top.
5. **Erasure analysis** (§7). A retention model separates the policies physically: tombstones retain removed payloads (visible-store correctness without erasure); excision and epochs erase; *address-only* tombstones support deletion audits at a

deniability cost quantified by payload min-entropy, with a dilemma between deduplication and deniable erasure (Remark 18).

6. **Verification and measurement** (§9, §8). The lattice-level core is kernel-checked in the same self-contained constructive Lean 4 development as the growth theory; survivor stability and product monotonicity are axiom-free. A 12-item machine ledger and 600 randomized property instances check everything including the derivation level. Cone excision beats epoch re-closure by 4.8–$8.3\times$ at bit-identical fingerprints; corpus-scale deletion analytics quantify the materialized-vs-complete gap and the anomaly frequency.

All artifacts — reference implementation, property suite, benchmarks, and the extended Lean file — accompany this paper; every number below that is not a citation is produced by them.

2 Preliminaries

We use the setting of [12], summarized for self-containment. The universe U holds typed parts; an *atom* carries an ingested payload and a descriptor; composites are generated by four rules (bind: $m \geq 3$ atoms sharing a descriptor, gated by an MDL code-length test; lift, frame, close assemble the upper levels). The address of a part is a SHA-256 hash of its kind, rule, sign, and the sorted list of its children's addresses; an atom's address hashes its payload. The standing hypotheses:

- **H1** (finitary) each rule instance has finitely many premises;
- **H2** (payload-determined) whether an instance fires is a function of its premises' payloads only;
- **H3** (stratified) generation strictly ascends the tower; no cycles;
- **H4** (injective addressing) the address function is injective on parts (collision resistance of SHA-256, the single extra-mathematical assumption of both papers).

Φ denotes the one-step operator and $\mathsf{K}(S)$ its least fixpoint over seed S. From [12] we use: K is a closure operator (T1); *conservativity*: $S \subseteq S' \Rightarrow \mathsf{K}(S) \subseteq \mathsf{K}(S')$ with surviving parts at identical addresses (T2); *incrementality*: $\mathsf{K}(S \cup D) = \mathsf{K}(\mathsf{K}(S) \cup D)$, computable frontier-locally (T3–T4); *history independence* of the canonical fingerprint (T5); and the impossibility of payload-determined maximal parsimony (T11). Throughout, $S \setminus D$ is set difference on atoms with $D \subseteq S$.

3 Derivation structure

Definition 1 (Premise-inscribing rules)**.** A rule format is *premise-inscribing* if the address preimage of every generated part contains the addresses of all premises of the generating instance.

The four rules of the system are premise-inscribing by construction: bind writes the sorted kernel-atom addresses, lift and frame the single child, close the section list. This is not an implementation accident but the substance of Merkle identity: a part *is* (the hash of) its derivation interface.

Lemma 2 (Single derivation). *Under H2–H4 and premise-inscribing rules, every part $p \in \mathsf{K}(S)$ has exactly one derivation: a unique generating rule instance, hence (by induction along H3) a unique derivation tree whose leaves are atoms of S.*

Proof. Suppose instances ι_1, ι_2 both generate p. Each writes its rule tag and full premise address list into the address preimage (Definition 1); by H4 equal addresses force equal preimages, so ι_1 and ι_2 have the same rule and the same premises, i.e. $\iota_1 = \iota_2$. Uniqueness of the tree follows by structural induction along the strict stratification H3, which terminates at atoms. $\square$

Definition 3 (Support; upward cone). For $p \in \mathsf{K}(S)$, $\mathrm{supp}(p) \subseteq S$ is the set of atoms at the leaves of p's unique derivation tree; for an atom a, $\mathrm{supp}(a) = \{a\}$. For $D \subseteq S$,

$$\mathrm{Up}(D) \;=\; \{\, p \in \mathsf{K}(S) : \mathrm{supp}(p) \cap D \neq \emptyset \,\}.$$

By Lemma 2, $\mathrm{supp}(p)$ is computed by traversing p's own child pointers — a payload-local operation requiring no global index and no provenance metadata. (Computing $\mathrm{Up}(D)$ over a *materialized* store benefits from an inverted child-to-parent index; the index is a cache, not content, and its absence changes cost, never semantics.)

4 The cone theorem

Theorem 4 (Cone theorem). *Under H1–H4 and premise-inscribing rules, for every seed S and $D \subseteq S$:*

$$\mathsf{K}(S \setminus D) \;=\; \mathsf{K}(S) \setminus \mathrm{Up}(D).$$

Proof. ($\subseteq$) Conservativity applied to $S \setminus D \subseteq S$ gives $\mathsf{K}(S \setminus D) \subseteq \mathsf{K}(S)$. Let $p \in \mathsf{K}(S \setminus D)$. Its unique derivation tree (Lemma 2, applied in the store over $S \setminus D$) has leaves in $S \setminus D$; by H4 this is the same tree as p's derivation in $\mathsf{K}(S)$ — the address pins the tree — so $\mathrm{supp}(p) \subseteq S \setminus D$, i.e. $p \notin \mathrm{Up}(D)$.

($\supseteq$) Let $p \in \mathsf{K}(S)$ with $\mathrm{supp}(p) \cap D = \emptyset$. Every node q of p's derivation tree satisfies $\mathrm{supp}(q) \subseteq \mathrm{supp}(p) \subseteq S \setminus D$. By induction along H3: the leaves are atoms of $S \setminus D$; each internal node's generating instance has all premises already established in $\mathsf{K}(S \setminus D)$ by the induction hypothesis, and fires there by H2 (firing depends only on premise payloads, which are unchanged). Hence $p \in \mathsf{K}(S \setminus D)$. $\square$

Corollary 5 (Survivor stability). *Deletion never disturbs survivors: every part of $\mathsf{K}(S \setminus D)$ exists in $\mathsf{K}(S)$ with the identical address, payload, and child list. The damage of a deletion is confined, definitionally, to the cone.*

Proof. $\mathsf{K}(S \setminus D) \subseteq \mathsf{K}(S)$ by Theorem 4; identity of records is H4 plus the fact that addresses pin preimages. $\square$

Corollary 6 (Semi-naive deletion). *$\mathsf{K}(S \setminus D)$ is computable from the materialized $\mathsf{K}(S)$ by removing $\mathrm{Up}(D)$ — one support pass, no re-closure, no rederivation check. Epoch re-closure (recomputing $\mathsf{K}(S \setminus D)$ from scratch) and cone excision produce bit-identical canonical fingerprints.*

Remark 7 (DRed collapses). Classical incremental view maintenance under deletion (DRed [5]) over-deletes everything derivable from the deleted facts and then *re-derives* what has an alternative derivation; counting variants maintain derivation multiplicities for the same reason. Lemma 2 makes the multiplicity structurally 1: in a premise-inscribing content-addressed closure there are no alternative derivations to re-derive, so over-deletion *is* exact deletion. The hard half of DRed is not optimized away here — it is absent by construction. (The price was paid upstream: a part derived two ways would be two parts, at two addresses; identity splits where provenance differs.)

5 What deletion costs

The complete field behaves perfectly under deletion — it only shrinks (Theorem 4 gives $\mathsf{K}(S \setminus D) \subseteq \mathsf{K}(S)$ directly). The parsimonious *views* of [12] do not.

Theorem 8 (Deletion anomaly). *Under the subsumption-quotient serialization (store the complete field; serialize only non-subsumed bindings), there exist S and $D \subseteq S$ with*

$$\mathrm{Ser}\big(\mathsf{K}(S \setminus D)\big) \not\subseteq \mathrm{Ser}\big(\mathsf{K}(S)\big) :$$

deleting atoms inserts serialized parts.

Proof. The witness of [12], run in reverse. Take the cohort $a_1, \ldots, a_4$ on one descriptor, $S = \{a_1, \ldots, a_4\}$, $D = \{a_4\}$. In $\mathsf{K}(S)$ the unique maximal binding is B_{1234}, so $B_{123} \notin \mathrm{Ser}(\mathsf{K}(S))$. In $\mathsf{K}(S \setminus D)$ the maximal binding is B_{123}, so $B_{123} \in \mathrm{Ser}(\mathsf{K}(S \setminus D))$. Executed on the reference implementation with the companion paper's constants, the inserted part materializes at address `362db034e4f1...` — precisely the address whose *retraction under growth* witnessed the impossibility theorem there. The same four atoms witness both directions of the trilemma. $\square$

The anomaly is not a corner case. On the natural corpus of §8, 58.1% of single-atom deletions trigger a view insertion: deletion inserts *more often than not*.

The anomaly admits an exact law. Write $C_c \subseteq S$ for the cohort of descriptor c, $n_c = |C_c|$, $k_c = |D \cap C_c|$, and $\tau = \max(3, m_0)$ for the admissibility threshold; recall that for $b > r$ the saving $\Delta(c, m)$ is strictly increasing in m (companion paper, Prop. 10), so a cohort of size $n \geq \tau$ has the *full-cohort binding* B_{C_c} as its unique $\subseteq$-maximal admissible binding, and the quotient serialization keeps exactly that binding's cone. Everything about when deletion inserts follows from this single observation.

Theorem 9 (Anomaly characterization). *Assume $b > r$ and the subsumption-quotient serialization* Ser. *For every seed S and $D \subseteq S$,*

$$\mathrm{Ser}\big(\mathsf{K}(S \setminus D)\big) \setminus \mathrm{Ser}\big(\mathsf{K}(S)\big) = \bigsqcup_{\substack{c:\, k_c \geq 1 \\ n_c - k_c \geq \tau}} \mathrm{cone}\big(B_{C_c \setminus D}\big),$$

where $\mathrm{cone}(B) = \{B, \ \mathsf{lift}(B), \ \mathsf{frame}(B)\}$*. In particular: (i) deletion inserts serialized parts* iff *some cohort is touched ($k_c \geq 1$) and remains bindable ($n_c - k_c \geq \tau$); (ii) insertions arrive in disjoint cones of exactly three parts, one per such cohort; (iii) the inserted binding's address is the content address of the surviving cohort, computable before the deletion is executed.*

Proof. Under $b > r$, every subset of size $m \in [\tau, n]$ of a cohort of size n is admissible and is strictly subsumed by the full cohort; cohorts of size $< \tau$ generate nothing. Hence $\mathrm{Ser}(\mathsf{K}(S))$ contains, per cohort with $n_c \geq \tau$, precisely $\mathrm{cone}(B_{C_c})$, and $\mathrm{Ser}(\mathsf{K}(S \setminus D))$ contains, per cohort with $n_c - k_c \geq \tau$, precisely $\mathrm{cone}(B_{C_c \setminus D})$; lift and frame are unary, so each kept binding contributes exactly its three-part cone, and cones at distinct descriptors are disjoint (addresses inscribe the descriptor; H4). Taking the difference at descriptor c: the new cone is absent from the old serialization iff $B_{C_c \setminus D} \neq B_{C_c}$, which by H4 (addresses pin premise sets) holds iff $k_c \geq 1$; and it exists iff $n_c - k_c \geq \tau$. Untouched or no-longer-bindable cohorts contribute nothing new. □

Corollary 10 (Single deletions; the anomaly frequency is a cohort-histogram functional)**.** *For a single-atom deletion $D = \{a\}$: insertion occurs iff $n_{c(a)} \geq \tau + 1 = \max(4, m_0 + 1)$, in which case exactly one cone of three parts is inserted. Consequently, deleting an atom chosen uniformly from S,*

$$\Pr[\textit{view insertion}] \;=\; \frac{1}{|S|} \sum_{c\,:\, n_c \geq \tau+1} n_c,$$

a closed-form functional of the cohort-size histogram alone — no closure need be computed. On the corpus of §8 ($\tau = 3$): atoms in cohorts of size ≥ 4 number 5014 of 8631, i.e. 58.1% — equal, to the reported precision, to the measured anomaly frequency. The measurement of §8 is Theorem 9 evaluated on the histogram.

Remark 11 (Scope)*.* The characterization uses $b > r$, which makes per-cohort maximality unique. For $b \leq r$, admissibility is not upward-closed in m and a cohort's maximal admissible bindings form an antichain of equal-size subsets; the analogous law holds with "the full-cohort binding" replaced by that antichain, at the cost of cones no longer being singletons per cohort. All measurements in this paper, and Prop. 10 of the companion paper, operate in the $b > r$ regime.

Theorem 12 (Deletion dichotomy)**.** *No payload-determined rule system, operating over the plain subset order on seeds, realizes a deletion semantics — i.e. a map from operation logs to stores under which removals strictly shrink the visible store on some input. Consequently any deletion-capable design must either (i) leave the payload-determined class at deletion points (*coordinate*: epoch barriers), or (ii) extend the order on states (*the two-phase product order of §6*).*

Proof. By the monotonicity theorem of [12] (mechanized there as the axiom-free `step_mono`), every payload-determined one-step operator is monotone, hence so is its closure: over the plain order, growing the ingested set can never shrink the store. A removal presented as ingestion (the only interface the plain order offers) therefore cannot decrease anything; the four-atom configuration above is a concrete input on which the required strict shrink fails. Option (i) abandons obliviousness exactly at deletion points — an epoch is a global agreement on a cut, i.e. coordination in the CALM sense [6, 2]; option (ii) changes what "growing" means, and §6 shows it suffices. The dichotomy is the deletion-side instance of the CALM boundary, in the same localized sense as the companion paper's Theorem 11: the boundary is CALM's; the content-addressed formulation — what coordination-freedom costs in *identity* and *retention* — is what this setting adds. □

6 The two-phase closure

Escape (ii) made precise. A *two-phase (2P) state* is a pair (A, R) of grow-only sets — adds and removes — ordered by the product of $\subseteq$ on A and $\supseteq$ on R; the *live store* is $\mathsf{K}(A \setminus R)$. Remove-wins is built into $\setminus$. This is the 2P-set of the CRDT literature [11] lifted from elements to closures: the novelty needed here is that the entire derived superstructure inherits the good behavior.

Theorem 13 (Product monotonicity)**.** *If $A \subseteq A'$ and $R' \subseteq R$ then $\mathsf{K}(A \setminus R) \subseteq \mathsf{K}(A' \setminus R')$. The live store is monotone over the 2P order; in particular every part of the live store keeps its exact address along any 2P-increasing history.*

Proof. $A \setminus R \subseteq A' \setminus R'$ pointwise (membership in A strengthens; non-membership in R weakens contravariantly), then conservativity. Mechanized axiom-free as `twoP_mono` (§9). □

Theorem 14 (Order-freedom under remove-wins)**.** *Let an operation log be a list of add-batches and remove-batches, and let the 2P semantics fold the log to $(A, R) = (\bigcup \textit{adds}, \bigcup \textit{removes})$. Then the live store depends only on the multiset of operations: any interleaving yields the identical canonical fingerprint. Without remove-wins this fails: under last-operation-wins state semantics, the two interleavings of $\{\textit{add } a, \textit{remove } a\}$ yield different stores.*

Proof. The fold is a commutative-monoid aggregation in each coordinate; mechanized as `live_swap` (segment transpositions, which generate all permutations, leave the live store unchanged — propext and quotient soundness only). The failure half is two executions: add-then-remove gives a dead in both semantics; remove-then-add gives a live under last-wins and dead under 2P (machine check DEL-11 exhibits the diverging fingerprints). □

Proposition 15 (Tombstone permanence)**.** *If $a \in R$ then $\mathsf{K}((A \cup \{a\}) \setminus R) = \mathsf{K}(A \setminus R)$: re-ingesting a removed atom is a no-op. Restoration of a removed atom under 2P therefore requires either an epoch reset or a fresh identity (a re-keyed payload), with the consequences of Remark 18.*

Proof. $(A \cup \{a\}) \setminus R = A \setminus R$ as sets when $a \in R$; apply K. Mechanized as `tombstone_permanence`. □

Deletion idempotence and commutation hold at the seed level for the same one-line reasons ($(S \setminus D) \setminus D = S \setminus D$; sequential differences merge to the union) and are mechanized as `deletion_idem` and `deletion_commute`; their store-level counterparts are checked at machine level (DEL-05, DEL-06).

7 Erasure: what is physically retained

Visible-store correctness and erasure are different properties. Define the *retention* of a policy as the set of removed-part payloads recoverable from the physical store after the deletion completes.

Proposition 16 (Retention separates the policies)**.** Epoch re-closure *and* cone excision *retain nothing of the cone: no removed payload survives the part records or the canonical serialization (machine check DEL-07 over randomized instances).* Tombstones *(2P with retained content) keep the visible store identical to excision — by Theorem 4 the close-then-filter and filter-then-close readings coincide — while retaining every removed payload: correctness without erasure, which is precisely the configuration that Art. 17 obligations [4] and the immutability–erasure tension [3] target.*

Definition 17 (Address-only tombstones)**.** Retain, for each removed part, its 64-hex address and nothing else; discard payloads and child lists. The retained address set supports a deletion *audit*: one can prove a specific part existed and was removed (its address re-derives from a claimed preimage) without retaining the part.

Remark 18 (The deniability boundary, stated honestly)*.* An address is a hash of content; preimage resistance [9] prevents inversion but not *verification of a guess*: an adversary holding a candidate payload can hash it and test membership in the retained address set. Address-only tombstones therefore provide erasure-with-deniability only up to the min-entropy of the removed payloads — a sentence from a published book is one dictionary lookup away. The standard fix, salting atom payloads with ingestion nonces, restores deniability and destroys content addressing's central economy: two ingestions of the same content no longer share an address. *Deduplication and deniable erasure pull against each other directly*; a store chooses, per corpus, which to keep. We flag this as a dilemma the formalism makes visible rather than one it resolves.

The policy space, summarized:

policy	visible store	coord.-free	survivor identity	erasure	audit
epoch re-closure	$\mathsf{K}(S \setminus D)$	no (barrier)	stable (Cor. 5)	yes	no
cone excision	$\mathsf{K}(S \setminus D)$	no (barrier)	stable	yes	no
tombstones (2P)	$\mathsf{K}(A \setminus R)$	yes (Thm. 13–14)	stable	**no**	yes
address-only 2P	$\mathsf{K}(A \setminus R)$	yes	stable	up to entropy	yes

One principle covers the table: *survivor identity is never the price of deletion — retention, coordination, and re-addability are.* Which of the three to pay is a policy decision the theory can now state exactly.

8 Empirics

Excision vs. epochs. On the data model of the companion benchmark (cohort-capped random corpora, cap 5; $|D| = 20$ random atoms; median of 5 runs; reference implementation, no index optimizations), cone excision of the materialized $\mathsf{K}(S)$ against epoch re-closure of $S \setminus D$:

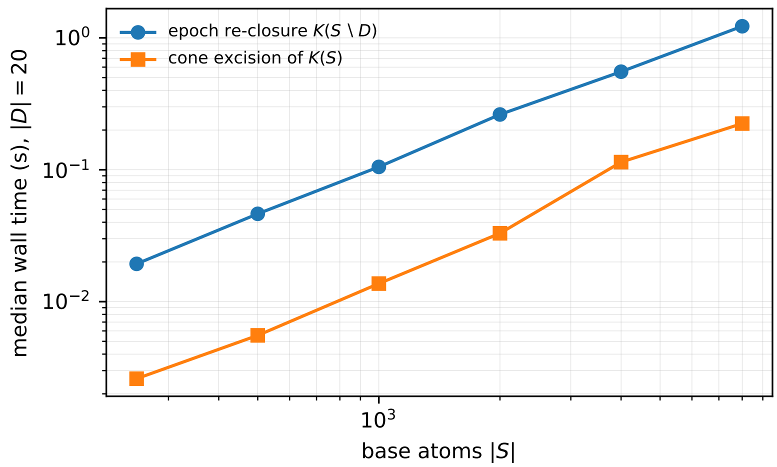

Figure 1: Median wall time, epoch re-closure vs. cone excision, $|D| = 20$, log–log. Fingerprint equality asserted at every point.

$\lvert S \rvert$	epoch (s)	excision (s)	speedup
250	0.0193	0.0026	$7.4\times$
500	0.0463	0.0055	$8.3\times$
1000	0.1051	0.0137	$7.7\times$
2000	0.2625	0.0330	$8.0\times$
4000	0.5541	0.1143	$4.8\times$
8000	1.2252	0.2240	$5.5\times$

Canonical fingerprints of the two procedures are asserted bit-identical at every size — the cone theorem, run live (Figure 1).

Corpus-scale deletion. On the Moby-Dick proxy of [12] (atoms = $8{,}631$ sentences; descriptor = most frequent non-stopword token; the re-fetched corpus reproduces the committed statistics exactly: $3{,}078$ descriptors, 637 bindable cohorts, maximal cohort $n = 268$), we analyze every possible single-atom deletion. The per-cohort mechanics are first verified on executed closures (a deletion inside a bindable cohort of size $n \geq 4$ removes exactly 4 materialized parts — atom, maximal block, section, root — and inserts exactly 3; at $n = 3$ the cohort dies: 4 removed, 0 inserted), then the exact integer combinatorics are evaluated over the cohort distribution:

- **Worst-case complete cone:** deleting one atom of the $n{=}268$ cohort places $\sim 10^{80.85}$ parts in $\mathrm{Up}(D)$ under complete semantics ($1 + 3\sum_{m=3}^{n}\binom{n-1}{m-1}$, exact integer evaluation) — against exactly 4 materialized parts under the quotient representation (mean 2.97, median 4, max 4 over all $8{,}631$ deletions). The cone is astronomical extensionally and constant-size materially; deletion cost is a property of the *representation*, with complete identity intact in both (every cone part has a well-defined address; H4 is a function, not a table). Figure 2 plots the gap.
- **Anomaly frequency:** $5{,}014$ of $8{,}631$ single-atom deletions (58.1%) trigger a view

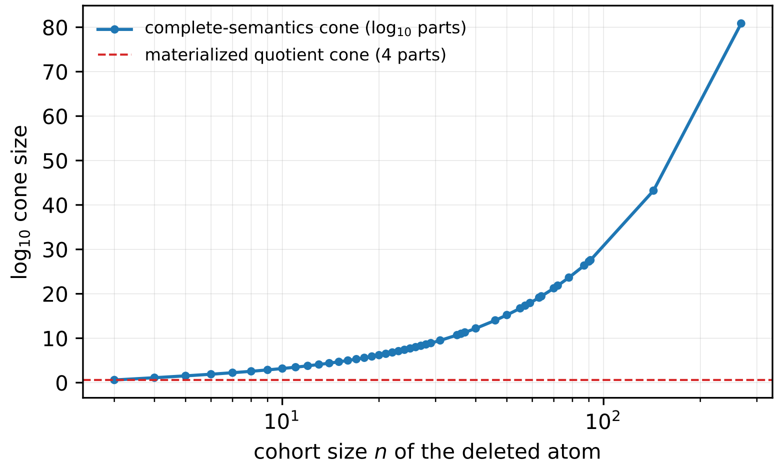

Figure 2: Single-atom deletion cone by cohort size on the corpus: complete-semantics cone ($\log_{10}$ parts, exact combinatorics) vs. the constant materialized quotient cone (4 parts).

insertion (the deleted atom sits in a bindable cohort of size ≥ 4). Theorem 8 is the common case, not the corner.

9 Verification

The claim ladder of [12], extended:

rung	artifact	checks
1	`delete_growth.py` (pure stdlib)	12-item ledger, 12/12 PASS
2	`test_delete_properties.py`	10 properties $\times$ 60 instances $=$ 600, all pass
3	`GrowthClosure.lean`, namespace `Deletion`	6 theorems, kernel-checked

The Lean development remains self-contained (no imports, no `mathlib`), fully constructive (no `Classical.choice` anywhere), and `sorry`-free. The compile-time axiom audit, verbatim:

```
'Growth.Deletion.survivor_stability' does not depend on any axioms
'Growth.Deletion.twoP_mono' does not depend on any axioms
'Growth.Deletion.deletion_idem' depends on axioms: [propext, Quot.sound]
'Growth.Deletion.deletion_commute' depends on axioms: [propext, Quot.sound]
'Growth.Deletion.tombstone_permanence' depends on axioms: [propext, Quot.sound]
'Growth.Deletion.live_swap' depends on axioms: [propext, Quot.sound]
```

Perimeter, stated exactly. The lattice level — survivor stability, 2P product monotonicity, deletion idempotence and commutation, tombstone permanence, remove-wins order-freedom — is kernel-checked above. The derivation level — Lemma 2 and the cone theorem, which require the premise-inscribing structure of the concrete rules — is proved on paper (§4) and machine-checked at rungs 1–2 (the cone theorem as bit-identical fingerprints of excision vs. epochs, deterministic and over 60 randomized instances). Mechanizing the derivation level would require formalizing the address algebra; we leave it as the natural next step of the formalization, not of the mathematics. As in the companion paper, collision resistance of SHA-256 (H4) is the single assumption outside the mathematics [9].

10 Related work

Incremental view maintenance under deletion. DRed [5] over-deletes and re-derives; counting methods track derivation multiplicities. §4 shows both collapse under premise-inscribing content addressing: multiplicity is structurally one, so over-deletion is exact (Remark 7). The cost was paid in identity — provenance-distinct derivations denote distinct parts.

CALM and coordination. The dichotomy of Theorem 12 is the deletion-side localization of the CALM boundary [6, 2]; the 2P escape is the lattice-extension move that the relaxation literature [13, 1] systematizes on the distribution side, here applied to the order on states. As in the companion paper, the boundary is claimed as CALM's; the identity-and-retention formulation is what the storage setting adds.

CRDTs. The 2P-set [11] and Merkle-CRDTs [10] supply the state design; the contribution here is closure-theoretic: the *derived* superstructure inherits monotonicity, confluence, and address stability (Theorems 13–14), with the inheritance kernel-checked.

Erasure and immutability. The regulatory tension between content-addressed immutability and Art. 17 erasure [4] is documented in [3]; §7 gives it an exact form — the retention model, the audit construction, and the entropy-bounded deniability of address-only tombstones [9] — and isolates the deduplication–deniability dilemma as the irreducible design choice.

11 Conclusion

Growth and deletion are not symmetric in a content-addressed closure, and the asymmetry is now exact. Growth is free: monotone, coordination-free, address-stable, frontier-local. Deletion costs exactly one of three currencies — retention (tombstones), coordination (epochs), or re-addability (2P permanence) — and never costs survivor identity. The cone theorem makes the deleted region definitionally minimal; the anomaly theorem shows parsimonious views pay in the opposite direction (insertion on deletion, 58.1% of the time on natural data); the dichotomy closes the door on rule-internal deletion. Open problems: identity-preserving restoration (re-add without epoch resets or re-keying); compaction of the grow-only remove set without losing order-freedom; and multi-tenant epochs, where different principals' erasure obligations cut the same store at different points — a coordination structure the present theory treats as a single barrier.

Artifacts. Reference implementation, property suite, benchmarks, corpus scripts, and the extended Lean development accompany the companion repository; companion paper: [12].

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